

Locomotion and Reproduction in Protozoa

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5.1 Locomotion in Protozoa

Though the protozoans are acellular and mostly microscopic, they are versatile in their locomotion. The free living aquatic protozoans move through the water, whereas some parasitic forms move through the body fluids of their hosts. Locomotion in protozoans is performed by cellular extensions such as **pseudopodia**, short hair-like organelles, the **cilia**, whip-like organelles, the **flagella** and the contractile fibrils, the **myonemes**.

5.1.1 Pseudopodia

These are found in **rhizopods**. The pseudopodia are temporary extensions of cytoplasm that develop in the direction of the movement. These temporary structures are useful to move on the substratum as our legs do, hence the name 'pseudopodia'. There are four kinds of pseudopodia namely **lobopodia** (blunt finger-like) as in *Amoeba* and *Entamoeba*, **filopodia** (fibre-like) as in *Euglypha*, **reticulopodia** (net-like) as in *Elphidium* and **axopodia** or **heliopodia** (sun ray-like) as in *Actinophrys*.

Pseudopodium is formed by the conversion of gel (viscous outer endoplasmic part) to sol (fluid-like inner endoplasmic part) and vice-versa. There are different theories on the formation of pseudopodium. The sol-gel transformation theory is the most accepted theory. Opinions differ on the site of contraction that pushes the pseudopodium forwards. Allen's theory of 'Front contraction' or 'Fountain zone' theory is

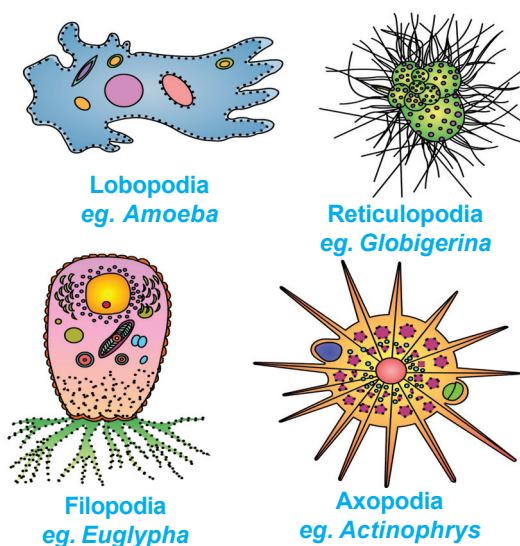


Fig. 5.1 Types of Pseudopodia

Plasmagel $\xrightarrow{+H_2O(\text{Solation})}$ Plasmasol $\xrightarrow{-H_2O(\text{Gelation})}$ Pseudopodium $\longrightarrow$ Displacement of the body in the forward direction

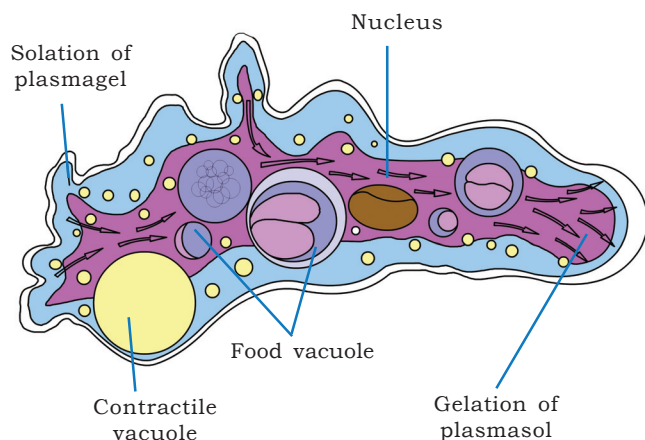


Fig. 5.2 Amoeba-locomotion

considered to be more appropriate. However modern research brings in the role of actin and myosin molecules too.

The pseudopodial or amoeboid movement is performed by **Amoeba**, **Entamoeba**, **Polystomella**, **Actinophrys**, etc. Amoeboid locomotion is probably the most primitive and the slowest type of locomotion.

In addition to protozoans, amoeboid locomotion is performed by amoeboid cells, macrophages, neutrophils, etc., of higher metazoans.

5.1.2 Flagella

The long whip-like locomotor organelles are called flagella (sing. flagellum). These are found in the **mastigophoran protozoans** (mastig – whip; phoran – bearer). Bacteria also possess flagella, which are structurally different from those of the eukaryotes. The movement of spermatozoa of animals involves flagellar movement.

I. Structure

The structural components of a typical flagellum include axoneme, microtubules, dynein arms, inner sheath, outer sheath, radial spokes, lateral appendages (such as mastigonemes or flimmers) and a basal granule (kinetosome).

- i. **Axoneme/ axial filament:** It is the central, longitudinal, microtubular structure of cilium and flagellum. It is surrounded by a membrane which is continuous with plasma membrane. All the components of the axoneme are embedded within the matrix.
- ii. **Microtubules:** An axoneme is made up of 2 central 'singlets' and 9 peripheral 'doublets (9+2 array)'. These are formed by the protein called **tubulin**. Each peripheral doublet consists of an outer 'A' and the inner 'B' tubules. So, the peripheral tubules are actually 9 microtubular doublets. (The microtubule 'A' is smaller but complete whereas 'B' is larger and incomplete). The peripheral doublets are interconnected by 'linkers' called **nexins**.
- iii. **Dynein arms:** The 'A' tubule of each peripheral doublet bears paired arms along its length, called dynein arms (Dyne - pulling like a dynamo). The dynein arms of 'A' tubule face the tubule 'B' of the adjacent doublet. They are oriented in the

same direction (clockwise) in all microtubules. The dynein arms are considered '**protein motor molecules**'. They are made of the protein, **dynein**.

- iv. **Inner and outer sheaths:** The two central singlets are enclosed by a fibrous inner sheath and the peripheral doublets are enclosed by an outer sheath (an extension of plasma membrane). The central singlets do not reach below the level of the pellicle or plasmalemma.
- v. **Radial spokes:** They are elastic fibres that connect the inner sheath, with the 'A' tubule of each doublet. They resemble the spokes that connect the rim of the wheel of a bicycle with the centre, hence the name 'radial spokes/ radial bridges'. The nine radial spokes limit the extent of sliding past of the doublets, during bending movements.
- vi. **Basal granule / kinetosome:** It is the cell organelle involved in the formation of a flagellum / cilium. The basal granule is a modified centriole. It is also called **kinetosome** [kineto - moving; soma - body] or **basal body** or **blepharoplast**. It is present in the ectoplasm. The basal granule is a cylindrical body, formed by '9' peripheral triplets arranged in a circle. The tubules of each triplet are designated A, B and C from the center towards the periphery. Both 'A' and 'B' cross the basal

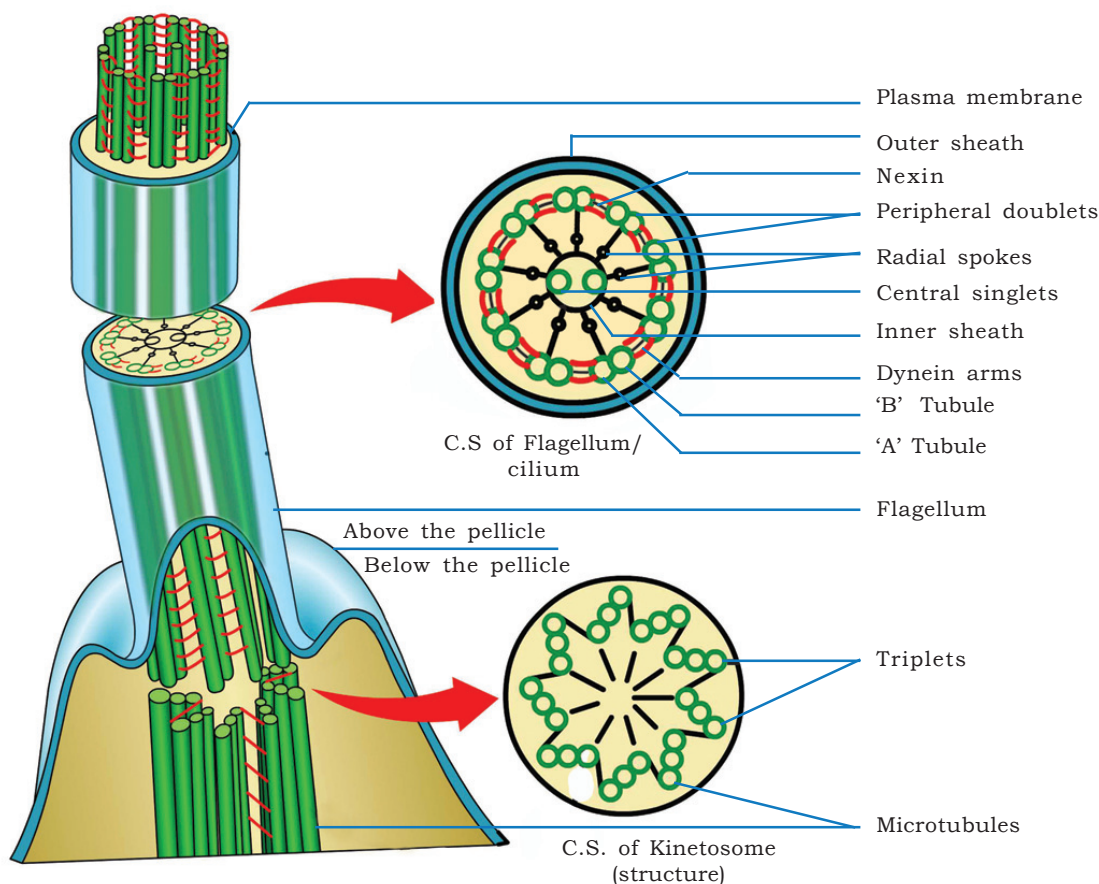


Fig. 5.3 Structure of a flagellum/ cilium

plate and continue as peripheral doublet above the pellicle in the axoneme. However, the tubule C terminates at the basal plate. Thus the 'triplets' of the basal granule are converted into the flagellar/ciliary 'doublets'. There are no central microtubules in the basal granule. The basal granule is also connected to the plasma membrane and nucleus by communication tubules, called **rootlets**. The rootlets can pull the flagellum and alter its orientation.

- vii. **Lateral appendages:** Some flagella bear one or two or many rows of short, lateral hair like fibrils called lateral appendages, also called '**mastigonemes**'.

II. Types of Flagella

Based on the presence or absence and/or the number of rows of lateral appendages, five types of flagella are recognised.

- Stichonematic:** This flagellum bears one row of lateral appendages on the axoneme. E.g. *Euglena* and *Astasia*.
- Pantonematic:** This flagellum has two or more rows of lateral appendages on the axoneme. E.g. *Peranema* and *Monas*.
- Acronematic:** This type of flagellum does not bear lateral appendages and the terminal part of the axoneme is naked without the outer sheath at its tip. E.g.: *Chlamydomonas* and *Polytoma*.

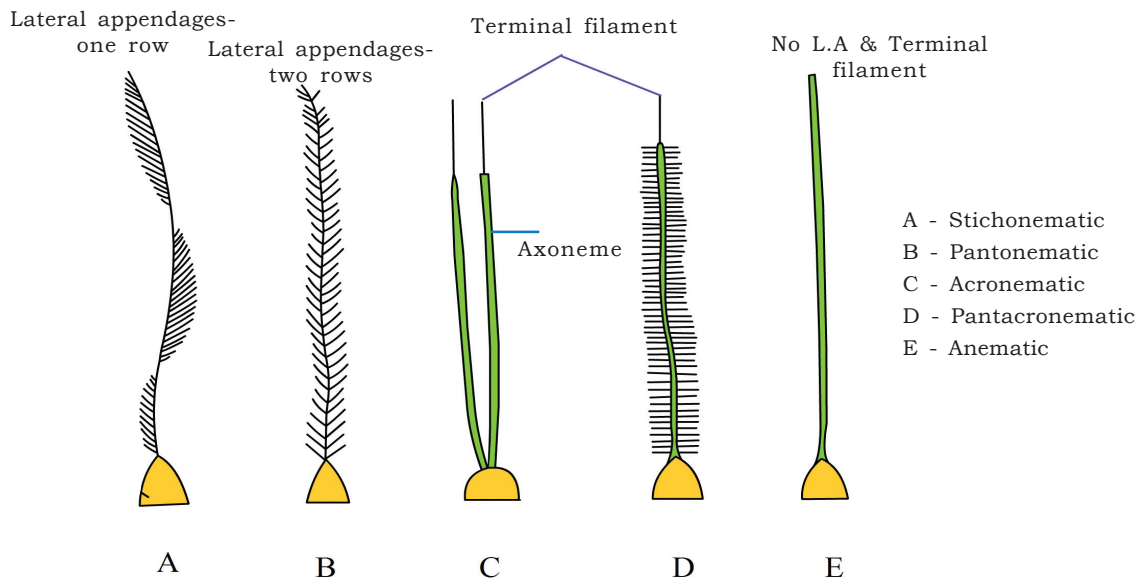


Fig. 5.4 Types of flagella

- d. **Pantacronematic:** This type of flagellum is provided with two or more rows of lateral appendages and the axoneme ends in a terminal naked filament. E.g. **Urceolus**.
- e. **Anematic or simple:** In this type of flagellum, lateral appendages and terminal filament are absent. Hence, it is called anematic (a-no; nematic-threads)
e.g. **Chilomonas** and **Cryptomonas**

III. Number of flagella

In the mastigophorans, the number of flagella varies from species to species. **Euglena** has two flagella (one long and one short), **Trypanosoma** has a single long flagellum arising at the rear end of the body. Four flagella are found in **Trichomonas**, four pairs in **Giardia** and many in **Trichonympha**.

5.1.3 Cilia

I. Structure

Cilia are small hair-like structures found in ciliate protists, epithelial lining of respiratory tract, genital ducts, ventricles of brain etc. In the primitive ciliates like **Paramecium**, the entire body is covered by cilia and in the advanced ciliates, such as **Vorticella**, cilia are restricted to the peristomial region. In the suctorians such as **Acineta**, the cilia are confined to the 'juvenile stages' only (The adults possess suctorial tentacles which help in feeding).

Cilia arise from the basal granules situated in the ectoplasm below the pellicle. Cilia are similar to flagella in their ultra structure. Cilia serve as organelles of locomotion, food collection and also act as sensory structures.

II. Infraciliary system

It is located just below the pellicle in the ectoplasm of a ciliate. It includes **kinetosomes**, **kinetodesmal fibrils** and **kinetodesmata**. The kinetosomes are present at the bases of cilia in transverse and longitudinal rows. The kinetodesmal fibrils are connected to the kinetosomes and run along the right side of each row of kinetosomes as a 'cord of fibres' called **kinetodesmata**. A longitudinal row of kinetosomes together with kinetodesmata constitute a unit called '**kinety**'.

All the kineties together form an infraciliary system, which is connected to a '**motorium**', located near the cytopharynx. The infraciliary system and motorium form the 'neuromotor system' that controls and coordinates the movement of cilia.

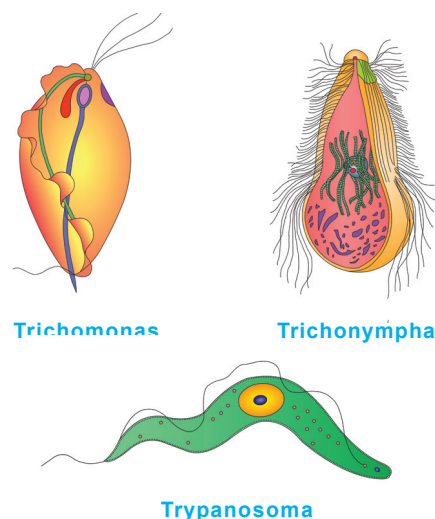


Fig. 5.5 Flagellates